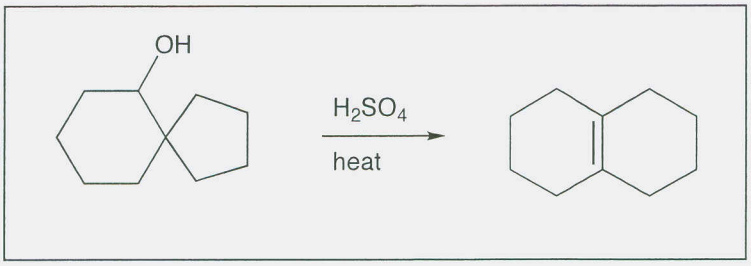
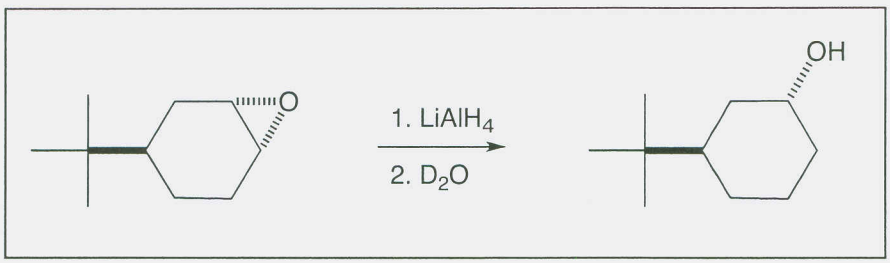
The following problems are all taken from Carey's Organic Chemisry, 7th Edition.

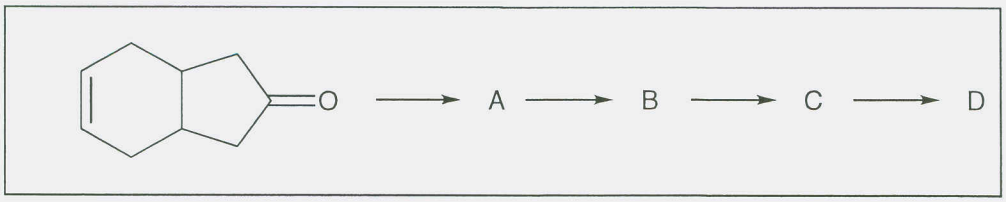
5.47(b). Draw the step-by-step electron pushing mechanism for the following reaction.



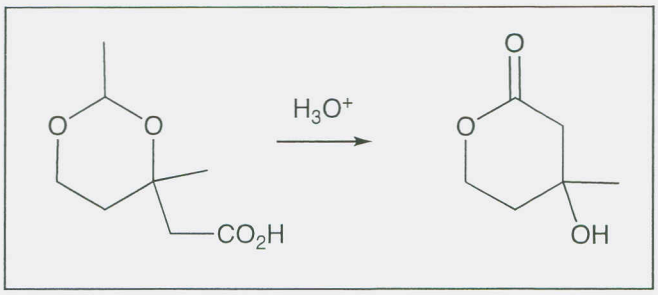
16.37. Given the reaction below, draw the product of the same reaction in its lowest energy chair conformation, but using $LiAlD_4$ instead of $LiAlH_4$.



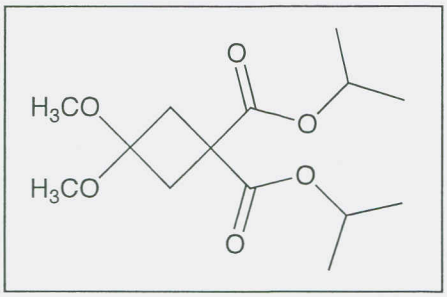
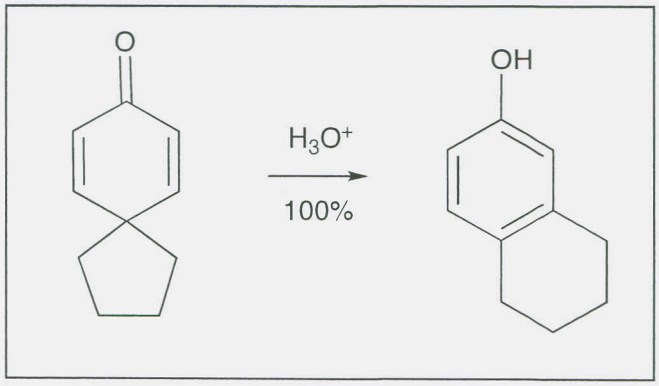
17.31. Wolff-Kishner reduction (hydrazine, KOH, ethylene glycol, 130 °C) of the compound shown gave compound A. Treatment of A with mCPBA gave compound B. Reduction of B with $LiAlH_4$ gave compound C. Oxidation of C with chromic acid gave compound D. Draw the structures of A-D (ignore stereochemistry).



19.29 On standing in dilute aqueous acid, the reactant below is smoothly converted to mevalonolactone. Draw a step-by-step electron pushing mechanism for the transformation.



21.27 (a) When the compound below was heated in refluxing aqueous HCl for 60 hours, a product with molecular formula $C_5H_6O_3$ was isolated in 97% yield. Draw the product. (b) Three other carbon containing products are formed in the reaction. What are they?



24.24. In a general reaction known as the dienone-phenol rearrangement, cyclohexadienones are converted to phenols under conditions of acid catalysis. Draw a step-by-step electron pushing mechanism for this reaction.